

Technical Document #247: Software Engineering - Comprehensive Overview

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Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Refactoring improves code structure without changing behavior. It makes code easier to understand, maintain, and extend.

Domain-driven design (DDD) models software around business domains. It improves communication between technical and business stakeholders.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Code review examines code changes before merging.
- API design defines how software components communicate.
- Domain-driven design (DDD) models software around business domains.